

§ 5.5.1

[Review] Theorem 5.34

How $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ can be implemented by a quantum circuit?

We use a circuit with ancilla (auxiliary) register.

Today's goal: Circuit which implement addition

In appendix B.

[Def B.2] Let $n \in \mathbb{N}$ and $a, b \in \mathbb{N}_0$ with $a, b < 2^n$.

$$a = \sum_{j=0}^{n-1} a_j 2^j, \quad b = \sum_{j=0}^{n-1} b_j 2^j, \quad a_j, b_j \in \{0, 1\}$$

We define $c_0^+ := 0$

$$c_j^+ := a_{j-1} b_{j-1} \oplus a_{j-1} c_{j-1}^+ \oplus b_{j-1} c_{j-1}^+, \quad 1 \leq j \leq n$$

$$s_j := a_j \oplus b_j \oplus c_j^+, \quad 0 \leq j \leq n-1$$

$$\text{Then, } a+b = \sum_{j=0}^{n-1} s_j 2^j + c_n^+ 2^n$$

Define operators on $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$,

$$A := \mathbb{I}^{\otimes 3} + (X - \mathbb{I}) \otimes |1\rangle\langle 1| \otimes \mathbb{I},$$

$$B := \mathbb{I}^{\otimes 3} + (X - \mathbb{I}) \otimes \mathbb{I} \otimes |1\rangle\langle 1|,$$

$$U_S := BA,$$

where X : $\mathbb{Q}$ -Not gate

Note $(|1\rangle\langle 1|)^* = |1\rangle\langle 1|$ and $(|1\rangle\langle 1|)^2 = |1\rangle\langle 1|$

$$X^* = X \quad \text{and} \quad 2(X - \mathbb{I}) + (X - \mathbb{I})^2 = 0$$

$\Rightarrow A$ is self-adjoint

$$A^2 = \mathbb{I}^{\otimes 3} + 2(X - \mathbb{I}) \otimes |1\rangle\langle 1| \otimes \mathbb{I} + (X - \mathbb{I})^2 \otimes |1\rangle\langle 1| \otimes \mathbb{I} = \mathbb{I}^{\otimes 3}$$

Hence, A is unitary.

In same way, B is unitary

U_S is unitary.

$$\text{Since } AB = BA \Rightarrow U_S^* = (BA)^* = AB = BA = U_S$$

$$A(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_1 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$B(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow U_S(|x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 \oplus x_1 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow U_S(|b_i\rangle \otimes |a_i\rangle \otimes |c_i^+\rangle) = |s_i\rangle \otimes |a_i\rangle \otimes |c_i^+\rangle$$

Define operators on $\mathbb{H}^{\otimes 4}$

unitary $C := \mathbb{I}^{\otimes 4} + (X - \mathbb{I}) \otimes |1\rangle\langle 1| \otimes |1\rangle\langle 1| \otimes \mathbb{I}$

self-adjoint $D := \mathbb{I}^{\otimes 4} + \mathbb{I} \otimes (X - \mathbb{I}) \otimes |1\rangle\langle 1| \otimes \mathbb{I}$

$$E := \mathbb{I}^{\otimes 4} + (X - \mathbb{I}) \otimes |1\rangle\langle 1| \otimes \mathbb{I} \otimes |1\rangle\langle 1|$$

$$U_C := EDC \text{ ; unitary.}$$

$$U_C^* = CDE$$

$$C(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_1 x_2 \oplus x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$D(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_3\rangle \otimes |x_1 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$E(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0 x_2 \oplus x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle$$

$$\Rightarrow \begin{cases} U_C(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |x_0(x_1 \oplus x_2) \oplus x_1 x_2 \oplus x_3\rangle \\ \quad \otimes |x_1 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle \\ U_C^*(|x_3\rangle \otimes |x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle) = |(x_0 \oplus x_1)x_2 \oplus x_1 \oplus x_3\rangle \\ \quad \otimes |x_1 \oplus x_2\rangle \otimes |x_1\rangle \otimes |x_0\rangle \end{cases}$$

$$U_c(|0\rangle \otimes |b_{j-1}\rangle \otimes |a_{j-1}\rangle \otimes |c_j^+\rangle)$$

$$= |c_{j-1}^+ (a_{j-1} \oplus b_{j-1}) \oplus a_{j-1} b_{j-1}\rangle \otimes |a_{j-1} \oplus b_{j-1}\rangle \otimes |a_{j-1}\rangle \otimes |c_j^+\rangle$$

$$\underbrace{= c_{j-1}^+ a_{j-1} \oplus c_{j-1}^+ b_{j-1}}_{= |c_j^+\rangle}$$

Def 5.36 $n \in \mathbb{N}$, $H^B := \overline{H^{\otimes n+1}}$, $H^A := \overline{H^{\otimes n}}$, $H^W := \overline{H^{\otimes n}}$

For $|b\rangle \otimes |a\rangle \otimes |w\rangle \in H^B \otimes H^A \otimes H^W$, we define

$$U_0(|b\rangle \otimes |a\rangle \otimes |w\rangle) := |b_n\rangle \otimes \bigotimes_{l=n_1}^0 (|b_l\rangle \otimes |a_l\rangle \otimes |w_l\rangle)$$

$$=: |\Psi[b, a, w]\rangle \in H^B \otimes H^A \otimes H^W$$

$$U_1 := \prod_{l=1}^{n-1} (\mathbb{1}^{\otimes 3l} \otimes U_c \otimes \mathbb{1}^{\otimes 3(n-l)})$$

$$U_2 := [(\mathbb{1} \otimes U_5)(\mathbb{1} \otimes \Lambda_{|1\rangle}(X) \otimes \mathbb{1}) U_c] \otimes \mathbb{1}^{\otimes 3(n-1)}$$

$$U_3 := \prod_{l=n-1}^1 (\mathbb{1}^{\otimes 3l} \otimes (\mathbb{1} \otimes U_5) U_c^* \otimes \mathbb{1}^{\otimes 3(n-l-1)})$$

$$U_+ := U_0^* U_3 U_2 U_1 U_0$$

Remark 1 In the proof of Thm 5.38, there will be precise explanation of operators.

$$\textcircled{2} \Lambda_{|1\rangle}(X) = \mathbb{1}^{\otimes 2} + (X - \mathbb{1}) \otimes |1\rangle \langle 1|$$

Lemma 5.37 U_0, U_1, U_2, U_3 and U_+ are unitary

pp) ① U_0 is unitary.

U_0 : change of bases $\Rightarrow U_0$ is unitary.

② U_1, U_3 are unitary

③ U_2 is unitary

$$U_2^* = [U_c^* (\mathbb{1} \otimes \underbrace{\Lambda_{|x\rangle}(x)^*}_{=\Lambda_{|x\rangle}(x)} \otimes \mathbb{1}) (\mathbb{1} \otimes \underbrace{U_s^*}_{=U_s})] \otimes \mathbb{1}^{\otimes 3(n-1)}$$

$$U_2^* U_2 = (U_c^* U_c) \otimes \mathbb{1}^{\otimes 3(n-1)} = \mathbb{1}^{\otimes 3n+1}$$

④ $\hat{U}_+$ is unitary

Thm 5.38 $\exists U_+$: circuit with ancilla on $H^{I/O} := H^B \otimes H^A$

For any $|\Phi\rangle \in H^{I/O}$

$$\hat{U}_+ (|\Phi\rangle \otimes |0\rangle^n) = (U_+ |\Phi\rangle) \otimes |0\rangle^n$$

Furthermore, for $a, b < 2^n$,

$$U_3 U_2 U_1 |\Psi[b, a, 0]\rangle = |\Psi[b+a, a, 0]\rangle$$

Rmk $|\Psi[b, a, 0]\rangle = U_0 (|b\rangle \otimes |a\rangle \otimes |0\rangle^n)$

$$\begin{aligned} \hat{U}_+ (|b\rangle \otimes |a\rangle \otimes |0\rangle^n) &= U_0^* |\Psi[b+a, a, 0]\rangle \\ &= |b+a\rangle \otimes |a\rangle \otimes |0\rangle^n \end{aligned}$$

On the other hand, $\hat{U}_+ (|b\rangle \otimes |a\rangle \otimes |0\rangle^n) = (U_+ (|b\rangle \otimes |a\rangle)) \otimes |0\rangle^n$

$$\Rightarrow U_+ (|b\rangle \otimes |a\rangle) = |b+a\rangle \otimes |a\rangle$$